25900560 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY202 - INTRODUCTION TO SPACE PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1.	Differentiate between aeronautics and astronautics with one example each.	[U] / [CO1]
2.	What is the heliosphere?	[K] / [CO6]
3.	Why space weather is monitored continuously ?	[U] / [CO4]
4.	What is limb darkening in photosphere?	[K] / [CO5]
5.	What is a geomagnetic storm ?	[U] / [CO2]
6.	State the typical speed range of fast and slow solar wind.	[K] / [CO2]
7.	What is meant by solar cycle?	[K] / [CO2]
8.	Explain one difference in orbit design between Aditya-L1 and Parker Solar Probe	[U] / [CO4]
9.	Define an ionogram.	[K] / [CO5]
10.	Draw a schematic of magnetosphere of Jupiter.	[U] / [CO3]
11.	What is scale height? Mention its significance.	[U] / [CO3]
12.	What is meant by frozen in condition of IMF?	[U] / [CO3]

13.	Explain recombination in the ionosphere.	[U] / [CO4]
14.	Explain how TEC cause ionospheric delay?	[U] / [CO5]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15.	What are the main characteristics of plasma and how it differs from solids, liquids, and gases?	[A] / [CO1]
16.	Compare the contributions of the Space Race era with the collaborative era of space exploration.	[Ap] / [CO6]
17.	What are coronal mass ejections (CMEs) and how do they differ from regular solar wind streams ?	[U] / [CO2]
18.	Describe the main objectives and instruments of the Solar Dynamics Observatory.	[U] / [CO4]
19.	Explain how the tipping-bucket toy model helps in understanding geomagnetic substorms.	[U] / [CO5]
20.	Explain the procecess occurring in the bow shock and magnetosheath.	[U] / [CO3]
21.	Explain the variation of different layers of earth’s ionosphere during night as compared to day.	[U] / [CO4]
22.	Explain how the solar wind interacts with Earth’s magnetic field to create the magnetosphere.	[U] / [CO5]

[5x6 = 30]